

AVNIR Energy – M2 projects

October 10, 2026

- Avnir Energy is an engineering company specialized in
  seismic, vibration, acoustics, ultrasound and ionizing radiations.

- Its domains go from Energy (nuclear, fossil and renewable) to Thermal, Hydraulic and Civil Engineering.

- It's current activity is mainly focused on **Nuclear Energy**
  - Seismic validation of nuclear power plants
  - γ-ray detector for nuclear fuel quality inspection (on-site prototype)

- Tools and Measure team:
  - Distribution of SYSCOM Instruments in France
  - On-site measures
  - Specific tool design

- Calculation team:
  - Simulations and qualifications for clients.

- R&D team:
  - New methods/tools conception

- We are exploring Structural Health Monitoring (SHM) methods:
  - Based on the **Time Reversal principle**
  - Where simulated and experimental data interweave
    1. Either numerical results for starting an experiment
       - Time reversal
    2. Or experimental data feeding a numerical method
       - Topological imaging

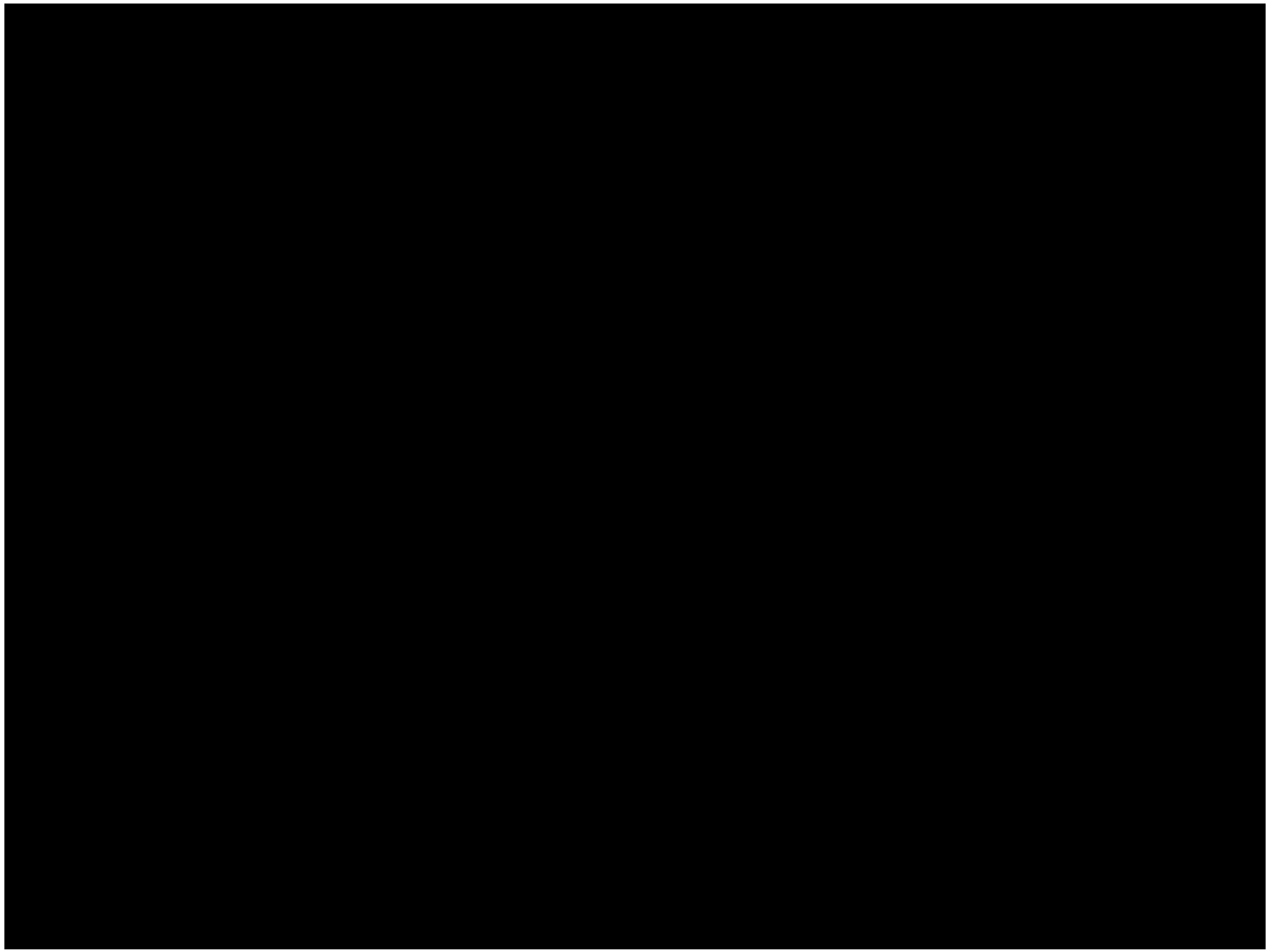
Project 1

Time reversal for nuclear tank monitoring



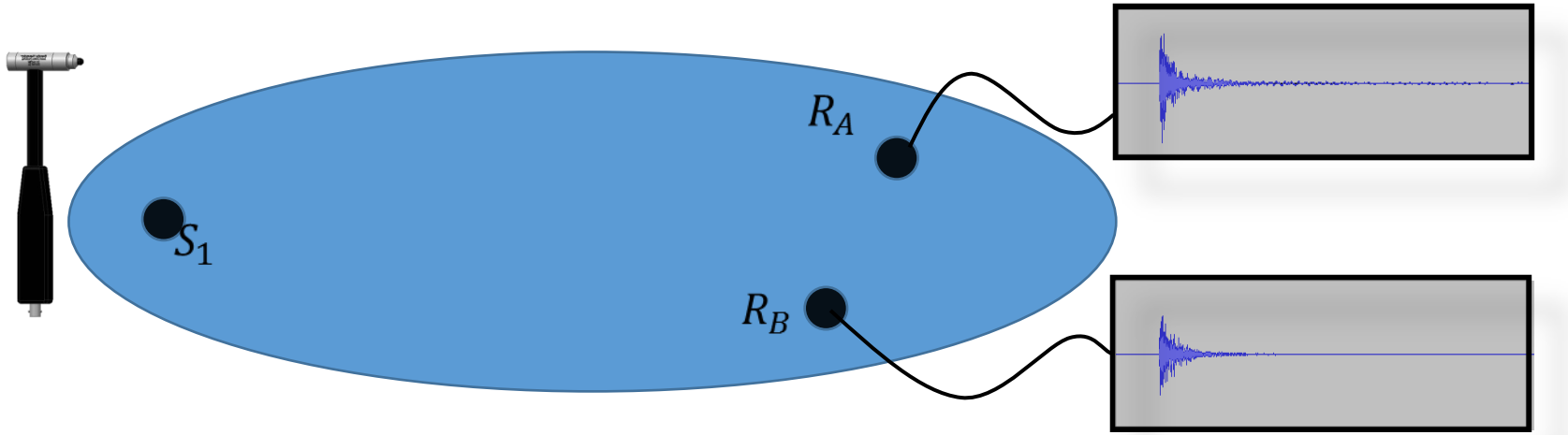
- Physical property :
  - The governing rules of vibration remain unchanged if the wave goes onwards or backwards

- This enables to **locate sources**

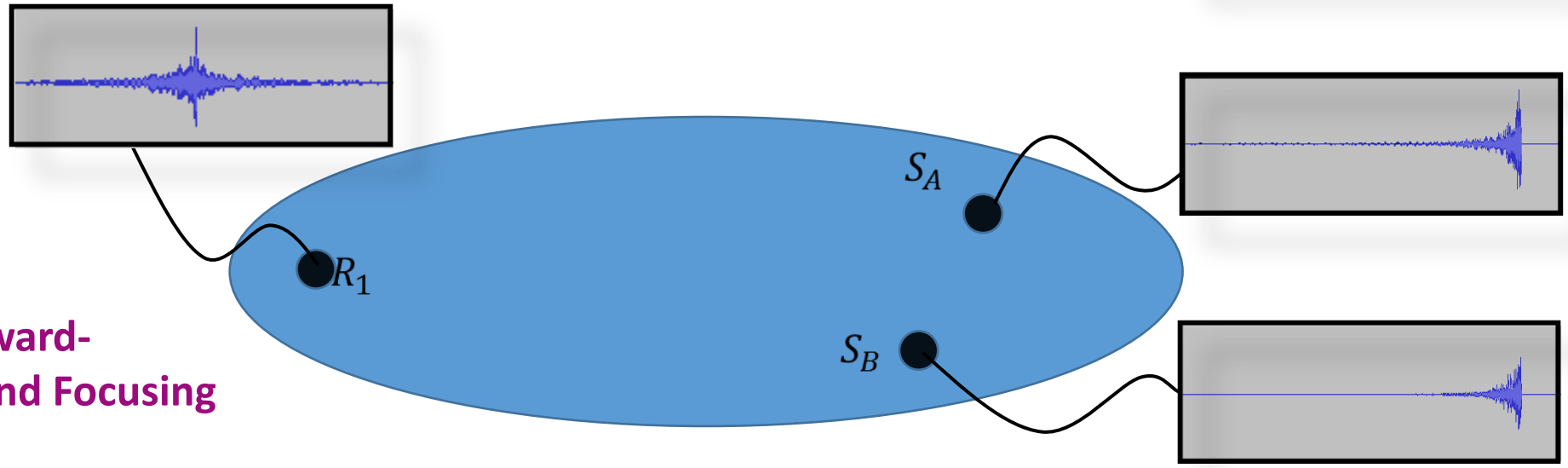


Experimental steps :

1- Recording

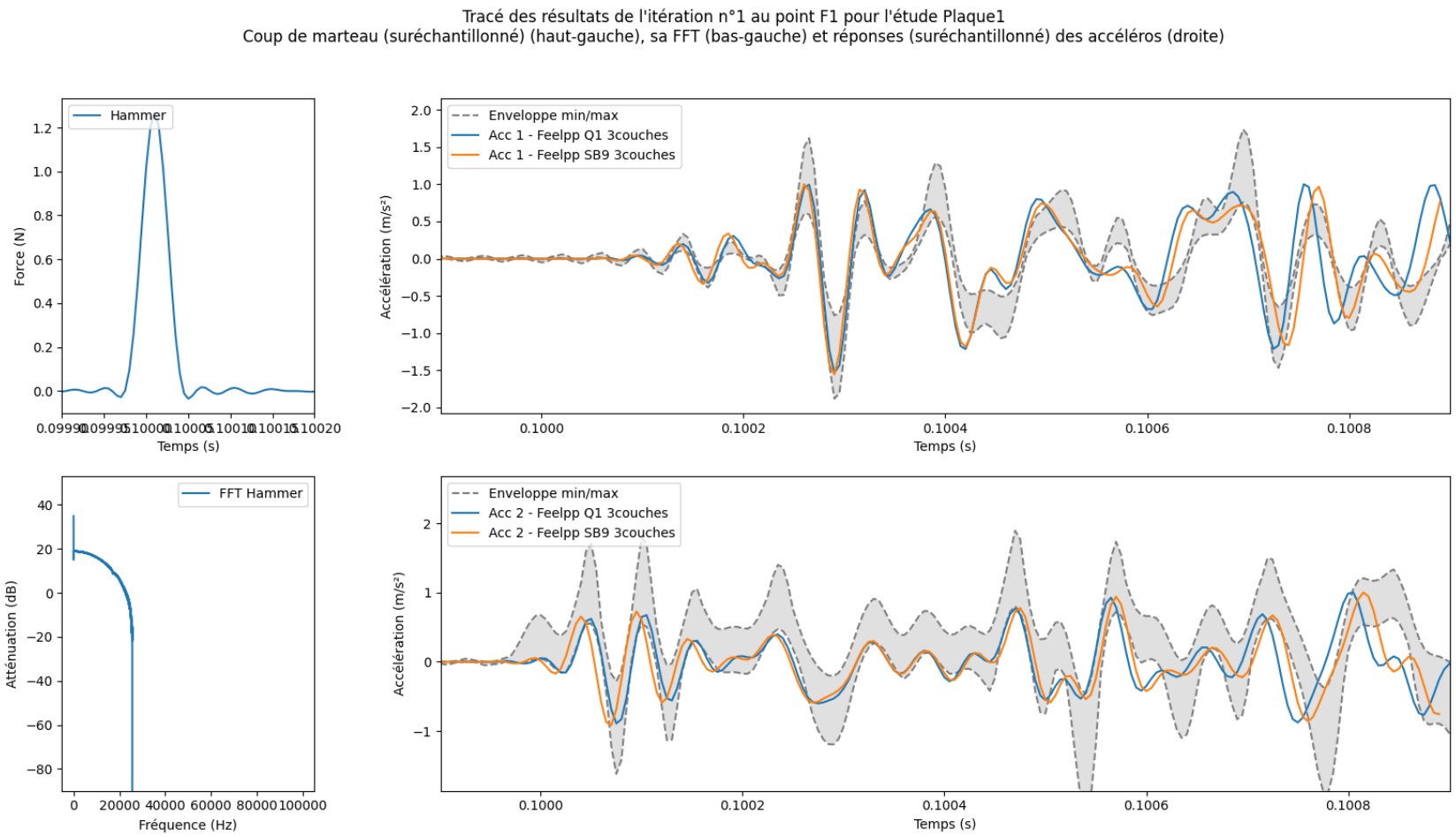


2- Wave backward-propagation and Focusing



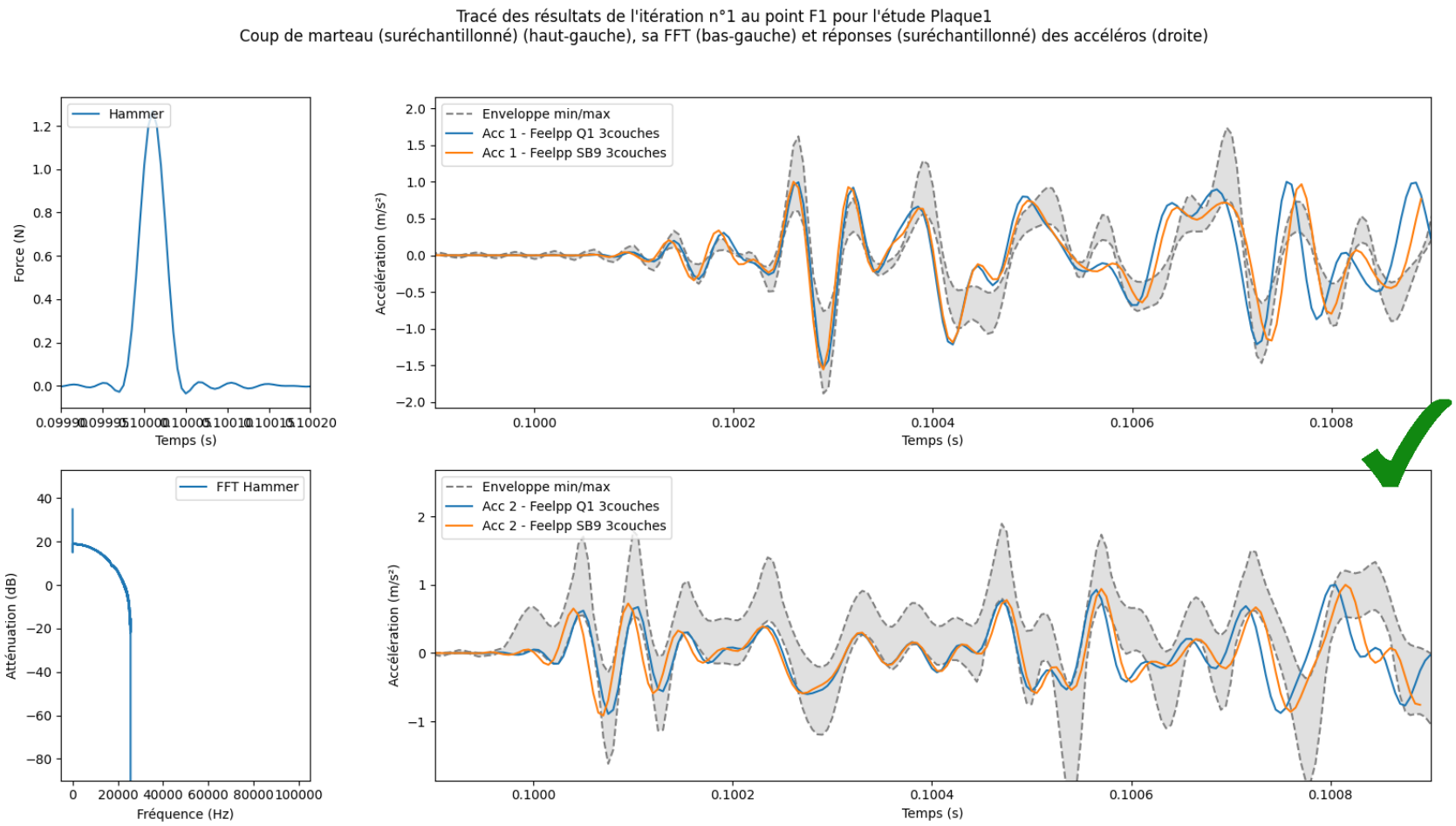
- Model the wave propagation in a simple medium : a small steel plate





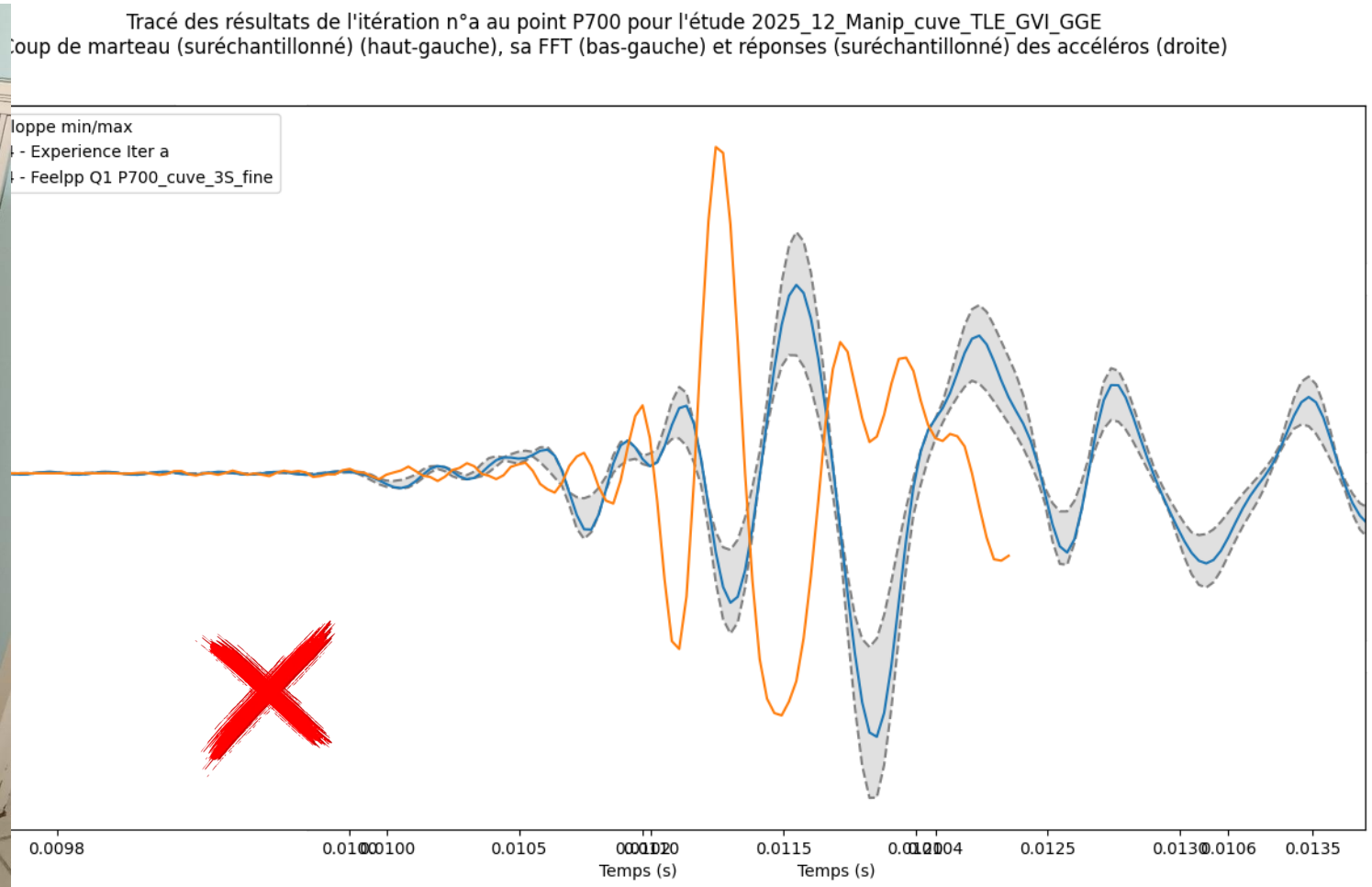
- Model the wave propagation in a simple medium : a small steel plate





- Model the wave propagation in a nuclear tank model





- **Hypothesis to verify:** Is the gravity no longer negligible ?
  - The tank mass is around 100 kg
  - And the steel plate is thin: 3mm

- Maybe the lower part of the tank is tighten as a drum skin ?
  - The higher the tension the higher the note
    - ➢ Because **waves travel at higher speed**

Project 1:

Add the gravity to the model and analyse the impact

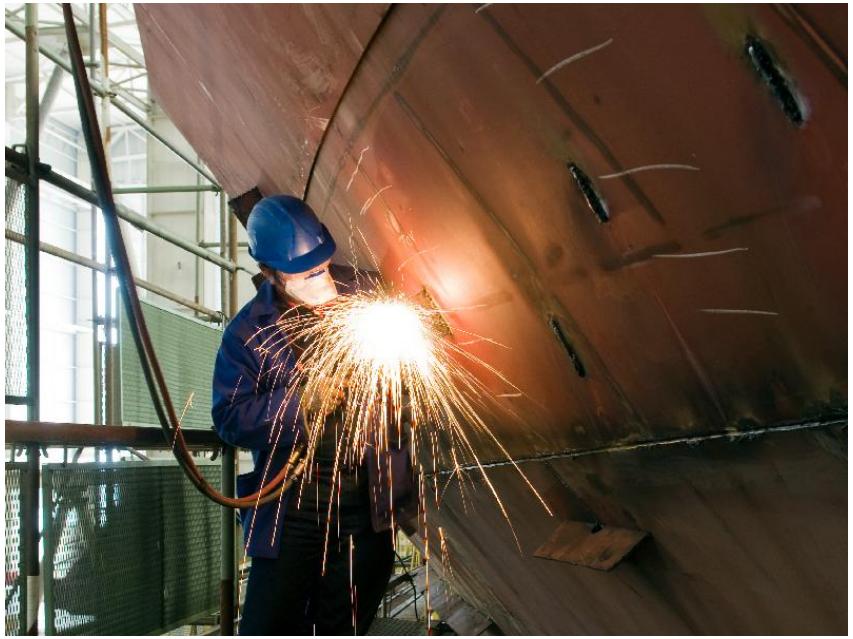


?

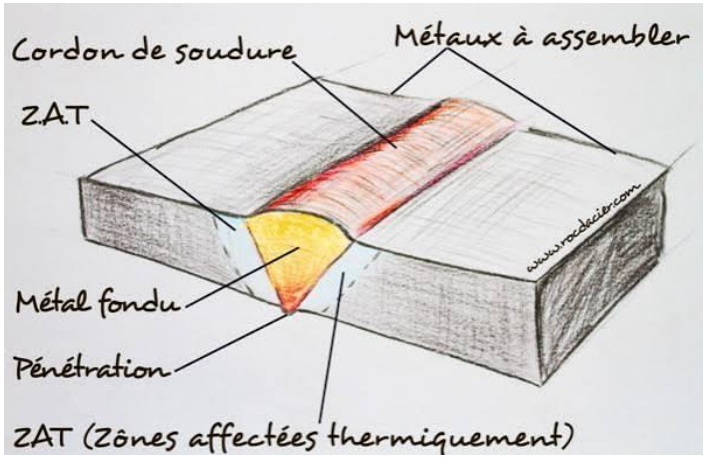


Project 2

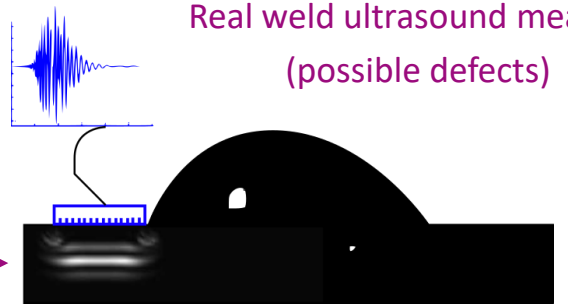
Weld quality monitoring





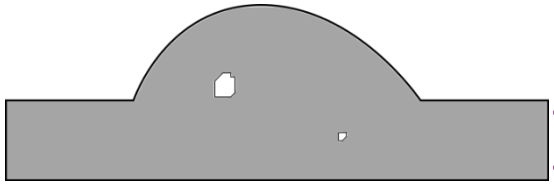


Real weld ultrasound measure
(possible defects)

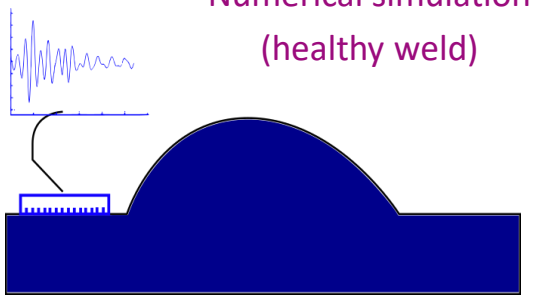


u_m

Weld to study (slice)



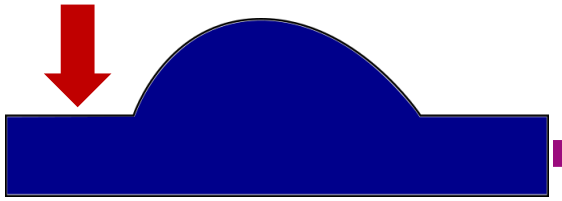
Numerical simulation
(healthy weld)



u_0

Healthy weld simulation
With **time reversed** source :

$(u_m - u_0)(T - t)$

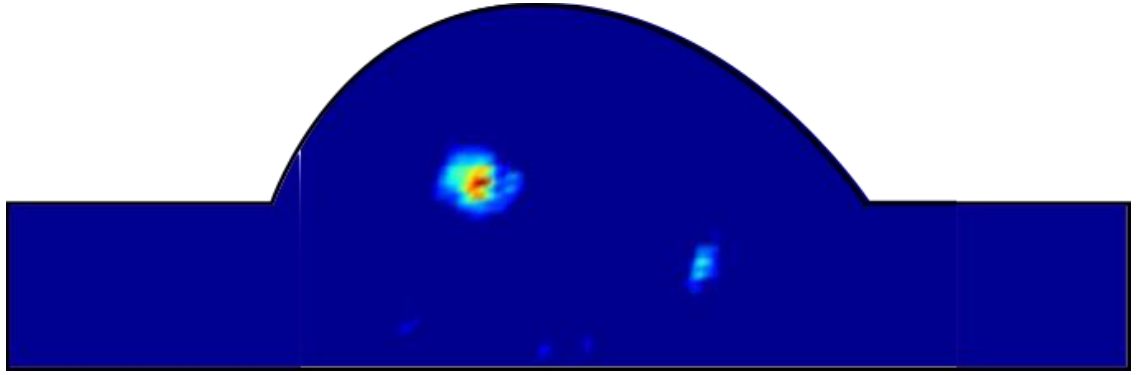


v_0

- Topological Energy computation

$$g_0(x) = \int_0^T \|U_0\|^2 \, \|V_0\|^2 dt$$

- Highlights the defects that act as secondary sources



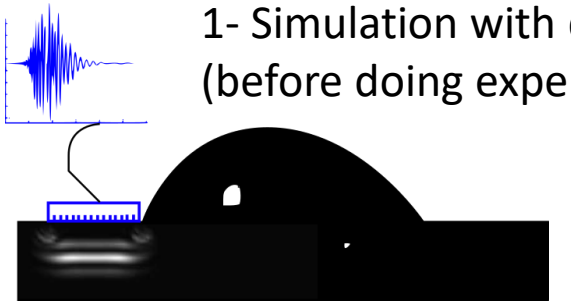
Project 2:

Build the all-steps program within Feel++ library

Challenges: Computation efficiency

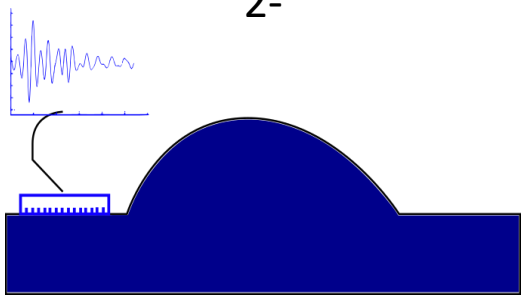
- ultrasound ➔ small grid and fine time steps
- Topological energy ➔ heavy cost

1- Simulation with defect
(before doing experiments)



➔ u_m

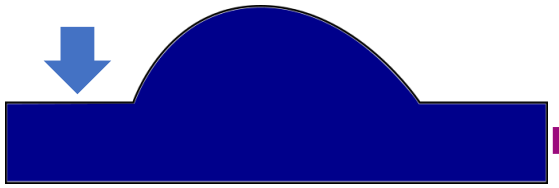
2-



➔ u_0

3-

$(u_m - u_0)(T - t)$



➔ v_0 ➔

4-

